Worksheet 12 - Inverse and Implicit Differentiation

MATH 1271

10/20/2026

1. For the following functions, determine their derivatives by recognizing them as inverse functions and applying inverse function rule.

   (a) $f(x) = \sqrt[n]{x}$.

   (b) $g(x) = \log_c(x)$.

   (c) $h(x) = \operatorname{arccot}(x)$.

2. For the following parts, let

$$f(x) = x^4 - 2x^3 + 3x^2 - 2x + 1.$$

(a) Compute $f(1)$.

(b) Use the previous part to compute $f^{-1}(1)$.

(c) What is the derivative of $f^{-1}(x)$ at $x = 1$?

(d) For which values of x is the line tangent to $y = f^{-1}(x)$ at $(x, f^{-1}(x))$ horizontal? For which values of x is it vertical?

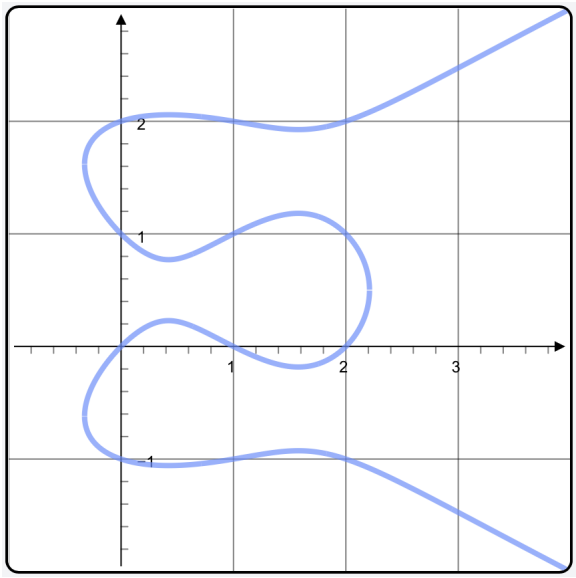
3. For each of the following equations, use implicit differentiation to find $\frac{dy}{dx}$, and determine the value of the derivative at the given point.

   (a) $x^3 - y^3 = 6xy$; $(-3, 3)$

   (b) $\sin(y) + y = x^3 + x$; $(0, 0)$

   (c) $e^{x/y} = x - y$; $(0, -1)$

4. Consider the curve defined by the equation $y(y^2-1)(y-2) = x(x-1)(x-2)$, whose graph is pictured below:



Through implicit differentiation, it can be shown that

$$\frac{dy}{dx} = \frac{(x-1)(x-2)+x(x-2)+x(x-1)}{(y^2-1)(y-2)+2y^2(y-2)+y(y^2-1)}$$

Use this fact to answer each of the following questions with 3 decimal points of accuracy.

(a) Determine all values of x at which the tangent line to the curve is horizontal.

Note: You may–and should!–use technology, such as wolframalpha.com, to find the needed zeros of the relevant polynomial function. For example, to have WolframAlpha solve the equation $x^2 - 3x + 1 = 0$ for you, type "Solve $x^2 - 3x + 1 = 0$" into the search box. It will output the answers $x = \frac{1}{2}(3-\sqrt{5})$ and $x = \frac{1}{2}(3+\sqrt{5})$.

(b) Determine all points (x, y) at which the tangent line to the curve is horizontal.

(c) Determine all points (x, y) at which the tangent line to the curve is vertical.

(d) Find the equation (in slope-intercept form) of the tangent line to the curve at one of the points where $x = 1$.